

DATA-DRIVEN PREDICTION AND ANALYSIS OF COOLING ENERGY USE IN TELECOM CENTRAL OFFICE ENVIRONMENTS

Anastasija Zubkova

M.Sc. student in Computer Science: Data Analytics and
Artificial Intelligence

Supervisor: Professor Dr. sc. ing. Dmitry Pavlyuk

Outline

- Telecom cooling problem and motivation
- Research gap and literature background
- Research aim, objectives, questions
- Intelligent control analytical framework, data and methods
- Forecasting and control results
- Interpretation, limitations, and research conclusions, Q&A

Problem and motivation

1. Operational necessity

Continuous cooling is required for reliable telecom operations



2. Efficiency opportunity

More intelligent cooling control could improve site-level energy efficiency



3. Practical gap

Telecom telemetry is mainly used for monitoring, alarms, and diagnosis, not for control-oriented decision support



4. Thesis motivation

Develop a benchmark-based and context-aware framework for intelligent cooling-control evaluation under real telemetry constraints

Research gap and literature background

- Telecom studies are relatively strong in monitoring, KPI design, and diagnosis
- Real cooling telemetry is often not directly suitable for control-oriented analysis and may require reconstruction of a defensible analytical basis
- Existing literature provides useful methods, but often assume ready targets, richer state feedback than are available

Research gap: no integrated telecom framework from target engineering to forecasting, benchmarking, and supervisory control under real telemetry constraints

Sorrentino et al. (2024); D'Aniello et al. (2018); Malafronte et al. (2021); Fan et al. (2021); Ji et al. (2017); Killian and Kozek (2016); Drgoňa et al. (2020); Xin et al. (2024)

Research aim and objectives

Aim: Develop and evaluate intelligent control strategies for cooling energy use in telecom central office environments using real monitoring data

Research tasks:

1. Review relevant literature
2. Prepare the control-oriented dataset
3. Develop the cooling-demand benchmark
4. Design context-aware control strategies
5. Evaluate control performance historically

Research questions

RQ1

To what extent can intelligent cooling control strategies identify opportunities to improve cooling energy use in telecom central office environments?

RQ2

Which forecasting approach provides the most reliable estimate of expected cooling demand for intelligent cooling control?

RQ3

Which operational and environmental variables are most relevant for constructing a control-oriented analytical basis for intelligent cooling control?

Object and subject of the research

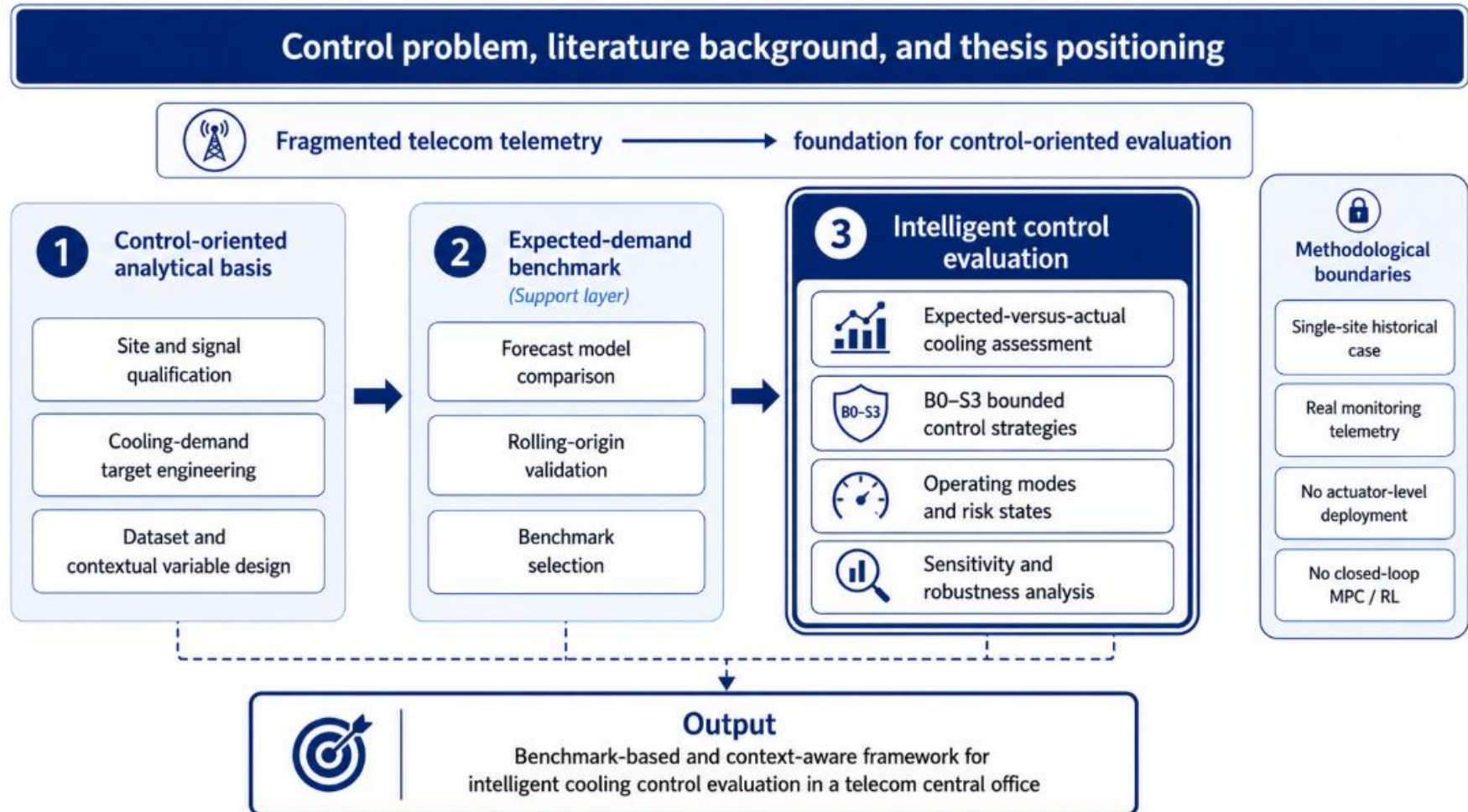
Research object

Cooling energy use in
telecom central office
technical infrastructure

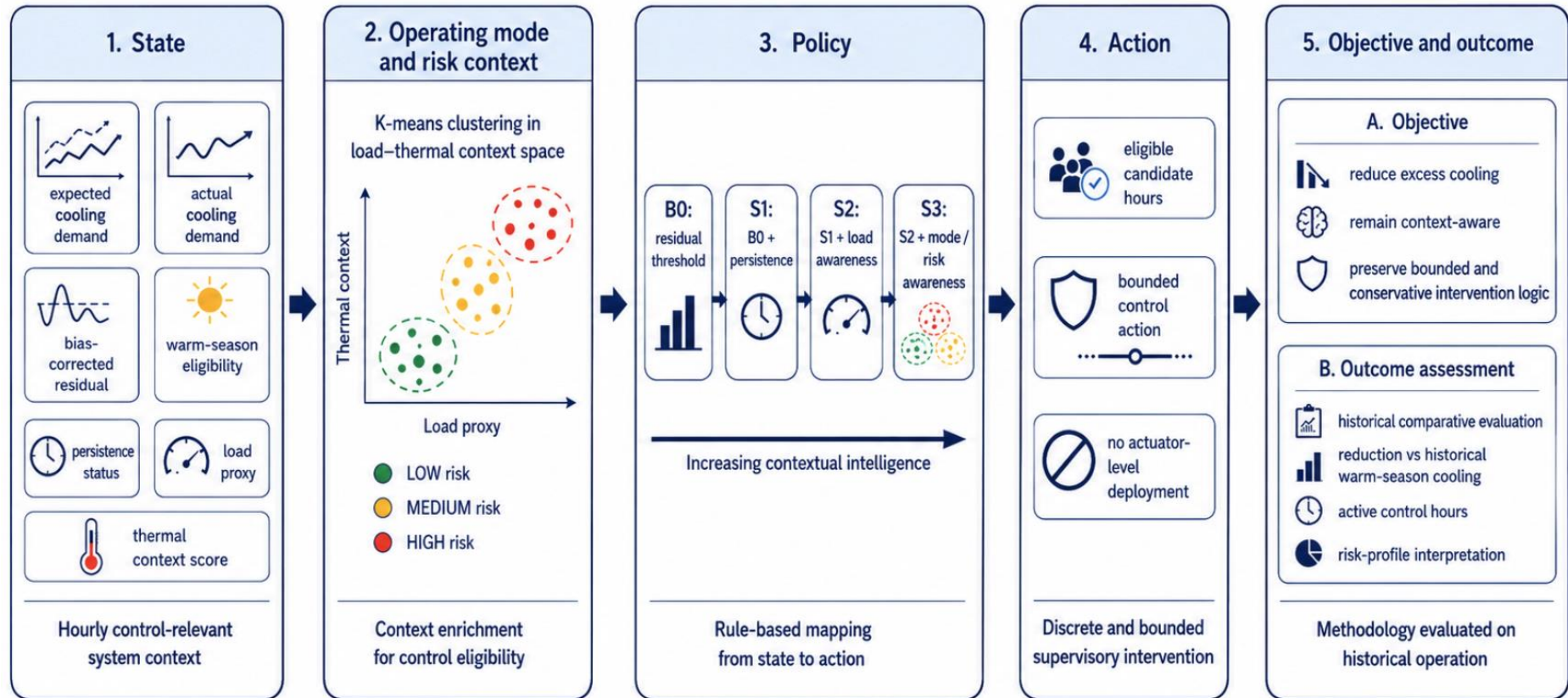
Research subject

Intelligent cooling control
in telecom central office
environments under real
operational telemetry
constraints

Analytical framework

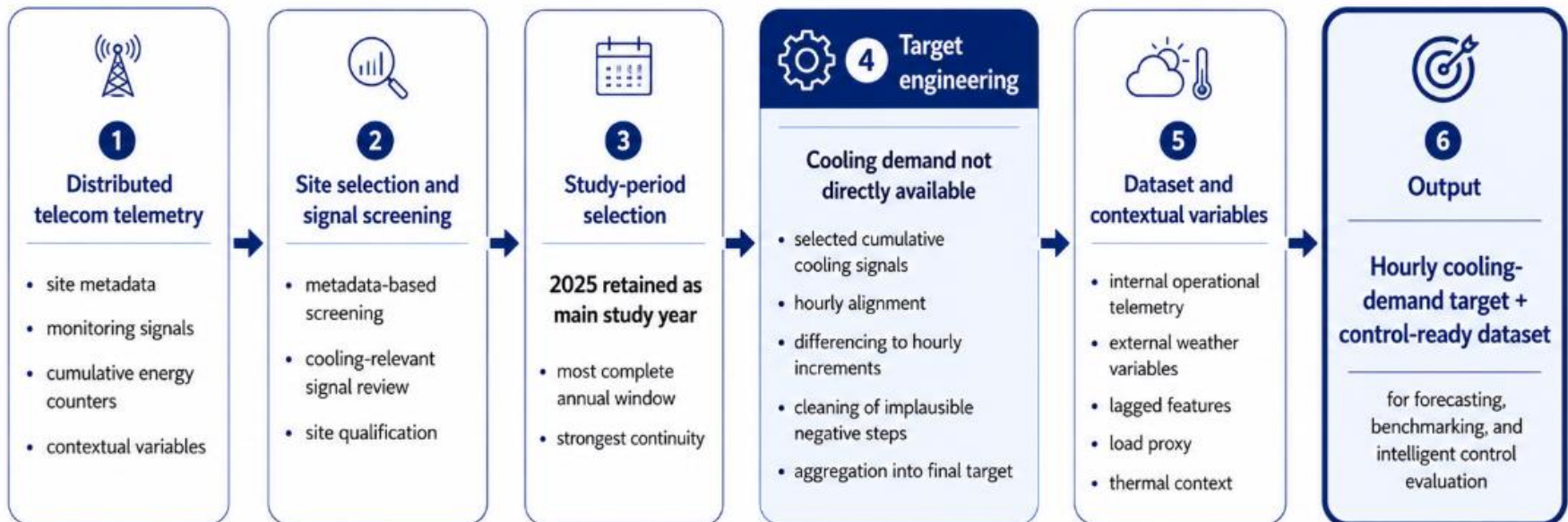


Intelligent control strategy



From telecom telemetry to a control-ready cooling dataset

- Cooling demand was not available as a ready-made hourly variable
- Site screening, cumulative-signal processing, and contextual variable design produced a control-ready dataset



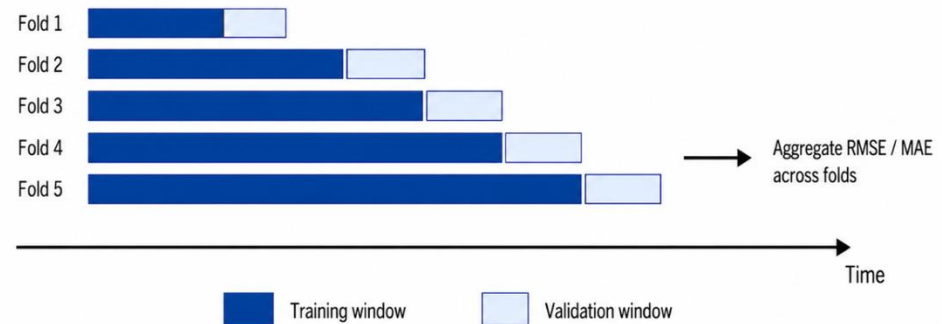
Forecasting and validation

- Models compared: naive, linear regression, RF, GB, SARIMAX, LSTM
- Final lag set: **1, 2, 3, 6, 12, 24, 48, 168 hours**
- Metrics: **MAE, RMSE, R^2**
- Validation: **chronological split + rolling-origin**

A. Initial chronological split

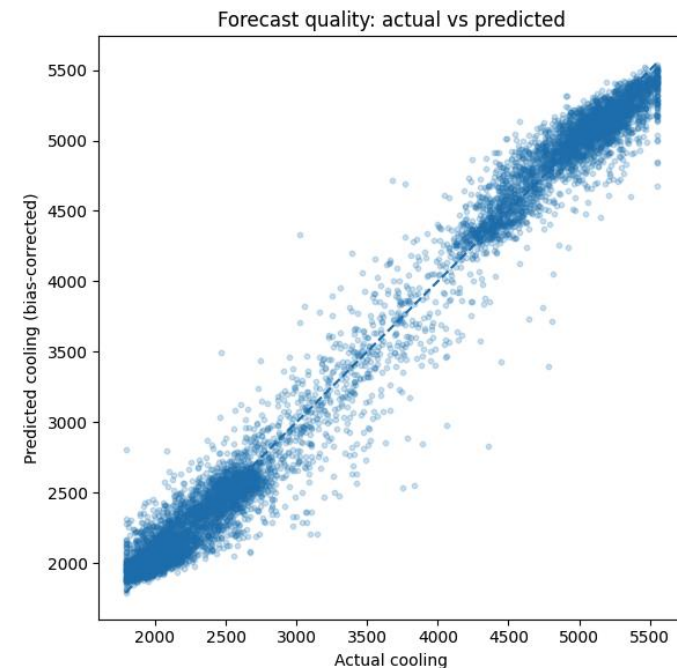
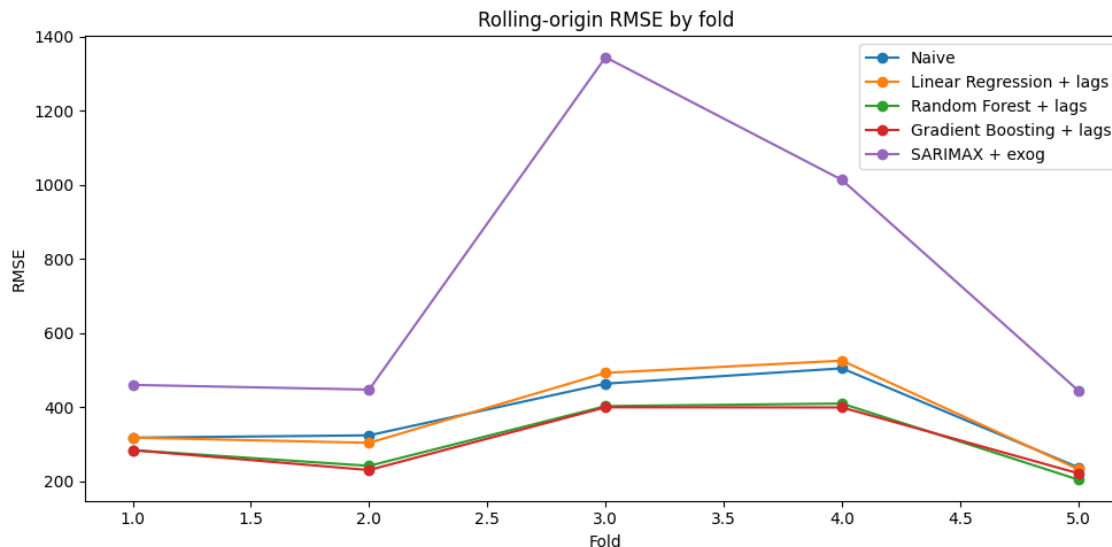


B. Rolling-origin validation



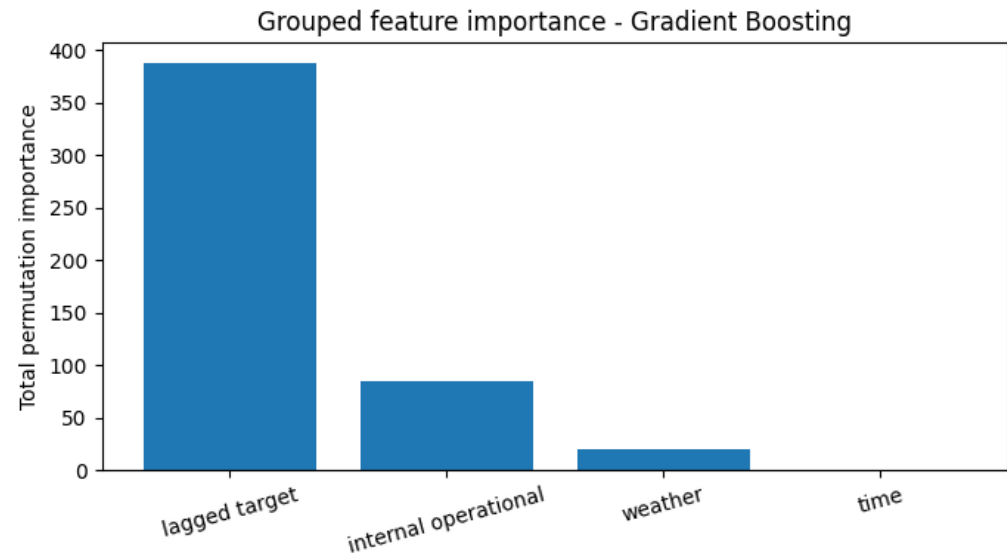
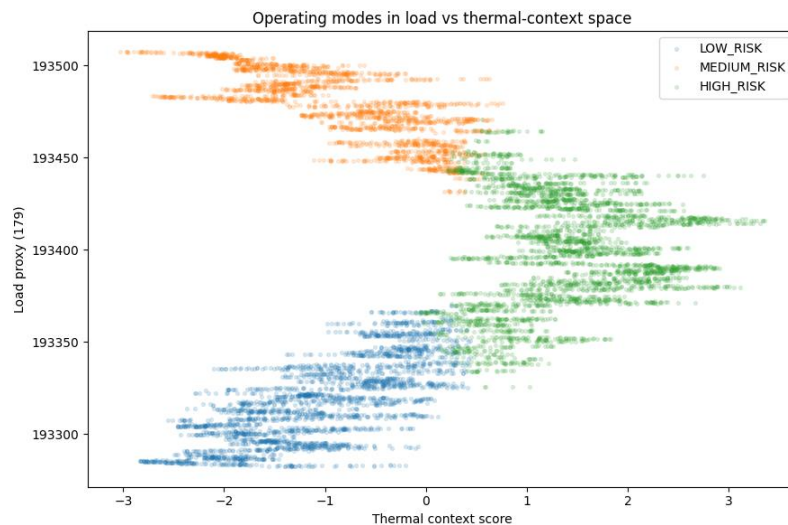
Expected demand benchmark selection

- **Gradient Boosting + lags** selected as final benchmark
- Strongest alternative: Random Forest + lags
- Best overall validation performance across rolling folds
- Final benchmark quality: **MAE = 115.06, RMSE = 161.79, $R^2 = 0.9855$**



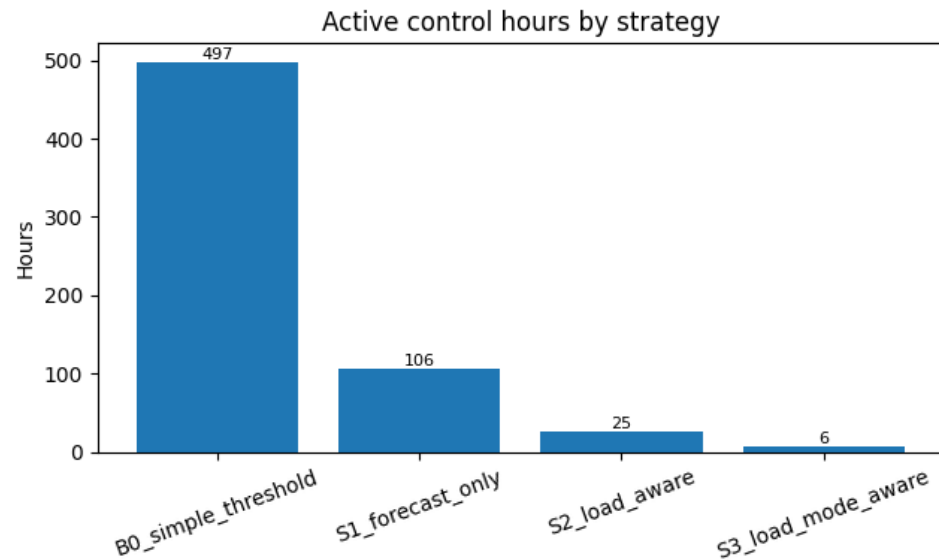
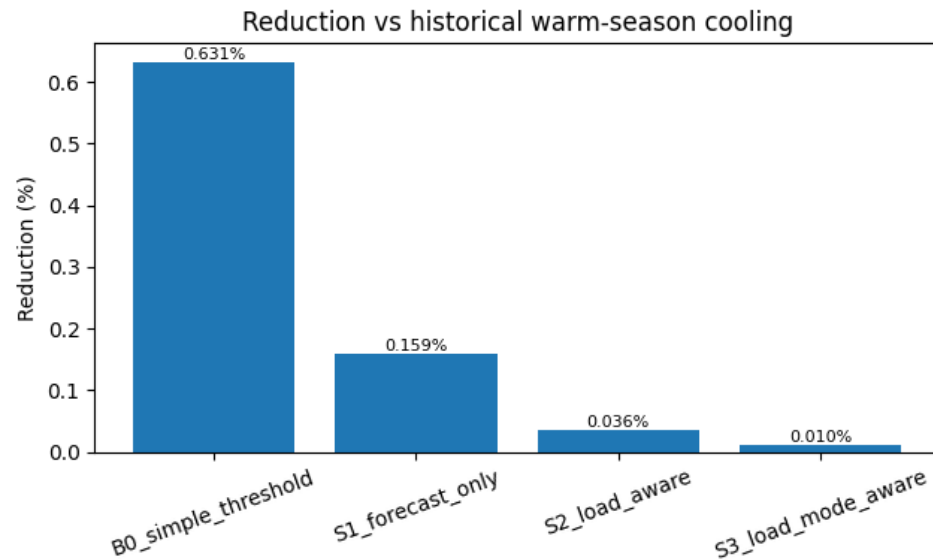
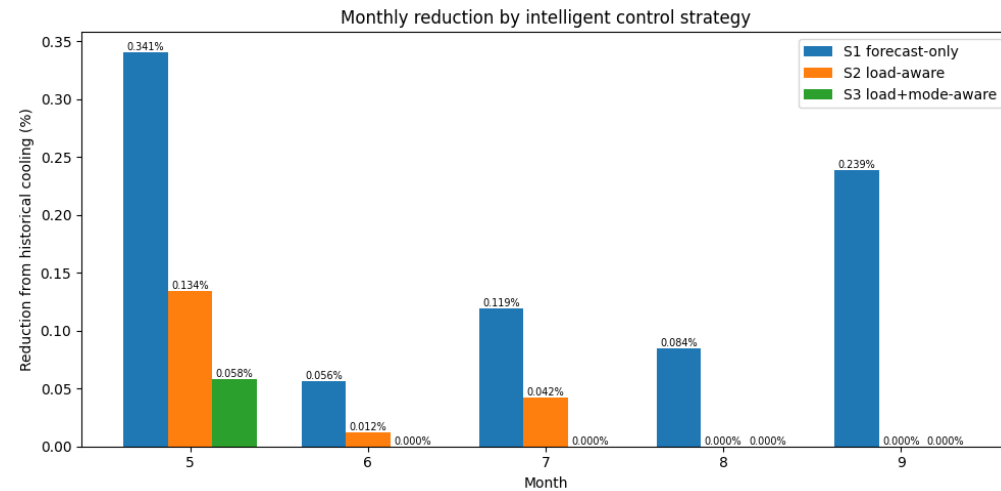
Most influential inputs and control-state basis

- Lagged cooling history was the strongest predictor group
- Short-horizon cooling demand was mainly persistence-driven
- Operational and weather variables refined the benchmark
- Load proxy and thermal context formed the control-state layer



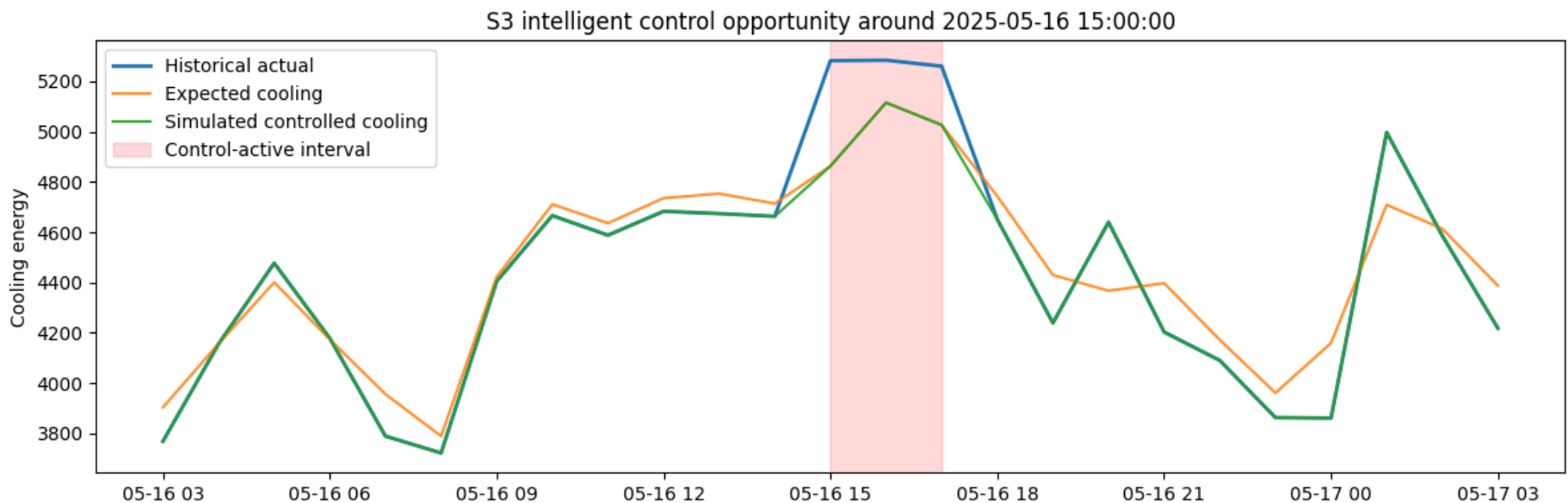
Comparative control outcome

- More context led to fewer active control hours
- B0 identified 497 hours, while S3 retained only 6
- Estimated reduction fell from 0.631% in B0 to 0.010% in S3



Example of control application

- Historical cooling exceeded the expected-demand benchmark
- Control was applied only when eligibility conditions were met
- Simulated cooling was reduced by a capped control action
- The 24-hour window shows persistence and activation context



Discussion

- The framework identified small, selective, and context-dependent control opportunities rather than large general savings
- The thesis supports energy-efficiency analytics through benchmark-based and context-aware cooling-control evaluation
- The study highlights a broader data-quality issue: credible control analytics depend on a defensible target and traceable state construction

Limitations

- Results depend on an engineered target, forecast benchmark and external weather only, no direct indoor thermal-state measurement
- The study uses one site and historical evaluation only; real-time and cross-site validation remain future work

Future work: multi-site validation, internal temperature data, validation for extreme weather conditions, and synthetic-data testing for limited-data conditions scenarios

Conclusions

- The thesis achieved its aim by showing that intelligent cooling control can be analysed in a telecom central office under real telemetry constraints
- Intelligent control was formulated as a bounded, benchmark-based, and context-aware analytical framework, not as direct live control
- The study linked target engineering, expected-demand benchmarking, and control evaluation into one traceable workflow from fragmented telemetry
- The main finding was that control opportunities existed, but they were small, selective, and strongly reduced by persistence, load, and risk-context filters
- The thesis supports energy-efficiency analytics by showing how potential cooling reductions can be assessed cautiously before any real deployment

Conclusions

- **RQ1:** Intelligent cooling control identified a small but defensible set of control opportunities, which became much narrower when persistence, load context, and risk-state conditions were added
- **RQ2:** The most reliable estimate of expected cooling demand was provided by a lagged nonlinear tree-based benchmark, with Gradient Boosting selected as the final model
- **RQ3:** The key inputs were the engineered cooling target, lagged demand history, and contextual state variables: external weather, load proxy, and thermal context
- The thesis supports a benchmark-based and context-aware approach to intelligent cooling control in telecom environments

Acknowledgements

The author would like to express sincere gratitude to **Dr.sc.ing., Professor Dmitry Pavlyuk** for supervision, academic guidance, and valuable feedback throughout this research.

The author also gratefully acknowledges **Swisscom** for providing the operational data used in this study, and extends special thanks to **Antons Lebedjko, Senior Software Engineer** and **Dirk Harberg, Project Manager** for their support in arranging data access and enabling this research.

Thank you, any questions?

Anastasija Zubkova

anastasijazubkovaa@gmail.com

Q1: What are the main limitations of relying purely on external weather telemetry instead of internal temperature branch signals when simulating control strategies for telecom central offices?

Root issue: external weather describes outside conditions, not indoor branch-level thermal state

Main limitations:

- 1. No local safety verification** → you cannot confirm whether reducing cooling would be safe near equipment branches.
- 2. Missed indoor variation** → external weather cannot show uneven temperatures between rooms, racks, or branches
- 3. No indoor thermal lag information** → indoor temperature may react later or differently than outside weather.
- 4. Limited deployment claim** → results remain benchmark-based simulation, not validated real-world control

<i>Q2: How would the proposed S3 context-aware strategy perform under extreme weather events not represented in the 2025 dataset?</i>	
Too-cold scenario	Too-hot scenario
• lower expected cooling • more apparent excess possible • risk of misreading abnormal operation	• higher thermal risk • fewer safe reduction hours • less reliable prediction outside observed conditions (extrapolation)
Future validation: • test multi-year hot/cold extremes • check against internal temperatures and incident records • start as alerts for manual review, not automatic control	
Summary: S3 would likely remain conservative, but hot and cold extremes may affect candidate hours differently and require validation beyond 2025	